Solar Cells

Solar cells (Photovoltaic devices) is made up of silicon that **converts sunlight directly into electricity**. Solar cell produces electricity without polluting the environment. Since it uses no fuel other than sunlight, no harmful gases, no burning and no wastes are produced. These can be installed in remote and inacessible areas (forests and hilly regions) where setting up of power plant is expensive.

Uses of Solar cells

(i) It can be used for street lighting, traffic signals, water pumping, battery charging system etc.
(ii) It is used in artificial satellites and space probes
(iii) It provides radio and TV transmission to remote areas
(iv) It is used in calculators, electronic toys and watches.

Solar Panel

Arrangement of many solar cells side by side connected to each other is called solar panel. The capacity to provide electric cuurent is much increased in the solar panel. But the process of manufacture is very expensive.



Figure 22.1 Solar Panel

Solar Cooker

It consist of an insulated metal box or wooden box which is painted from inside so as to absorb maximum solar radiations. A thick glass sheet forms the cover over the box. The reflector is the plane mirror which is attached to the box. The food is cooked by energy radiated by the sun.



Solar Cooker

Solar thermal power plant

In solar thermal power plants, many solar panels are used to concentrate sun rays, to heat up water into steam. The steam is used to run the turbines to produce electricity.



A capacity of 100 litres solar heater can save upto 1500 units of electricity per year.

Advantages of Solar Energy

(i) It is available in abundance in our country and is free of cost.
(ii) It is a renewable source of energy.
(iii) It can be used for generating electricity or heat.
(iv) It does not cause pollution.

22.6.2 Biogas

Biogas is the mixture of methane (nearly 75 %), hydrogen sulphide, carbon dioxide and hydrogen. It is produced by the **decomposition of animal wastes** (cow dung) and plant wastes in the absence of oxygen. It is also commonly called as '**Gobar gas**' since the starting material used is cow dung which means gobar in Hindi.

Uses of biogas

(i) It is used as fuel for cooking .
(ii) It is used to run motors and pump sets.
(iii) It is used to generate electricity.

Advantages of biogas

(i) It burns without smoke and therefore causes less pollution.
(ii) An excellent way to get rid of organic wastes like bio-waste and sewage material.
(iii) Left over slurry is a good manure rich in nitrogen and phosphorus
(iv) It is safe and convenient to use
(v) It can reduce the amount of greenhouse gases emitted.

10th_Science Unit-22.indd 320 12/3/2022 3:13:04 PM